

Foundation Models vs. Simple Baselines on Single-Cell Perturbation Prediction

An independent, transparent reproduction of an unresolved 2025 debate: does deep learning beat simple linear baselines at predicting post-perturbation gene expression, or is that conclusion an artifact of uncalibrated metrics? Three baselines, GEARS, and scGPT were benchmarked head-to-head on both datasets under conventional metrics *and* a calibration framework that scores each model against the gap between a no-signal negative control and a best-achievable positive control.

ADAMSON (SINGLE-GENE) scGPT wins both metrics, clearly — MSE@20DEG 0.124 vs. ~0.25 for the field	NORMAN (COMBINATORIAL) Split decision GEARS best on MSE@20DEG, ridge best on Pearson(Δ control)	CALIBRATION VERDICT No ranking flip calibrated fractions preserve the raw-metric ordering on both datasets	MODELS COMPARED 5 + 1 no-change, mean, ridge, GEARS, scGPT — Geneformer reported separately
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RESULTS AT A GLANCE

MODEL	MSE@20DEG	PEARSON(Δ CTRL)	CALIB. MSE FRAC.	CALIB. PEARSON FRAC.
Adamson — single-gene perturbations				
no_change	0.3800	—	0.0000	—
mean	0.2478	0.6357	0.3492	0.6741
ridge	0.2482	0.6357	0.3483	0.6741
GEARS	0.2465	0.5557	0.3526	0.5892
scGPT	0.1240	0.6405	0.5720	0.6710
Norman — combinatorial perturbations				
no_change	0.4368	—	0.0000	—
mean	0.3316	0.5786	0.2439	0.6427
ridge	0.2308	0.6732	0.4775	0.7478
GEARS	0.1689	0.5831	0.6210	0.6478
scGPT	0.2235	0.5505	0.5077	0.6239

Scope note. The positive control is built from repeated random 50/50 splits of each real test condition's own held-out cells (this dataset has no true biological-replicate metadata) — a documented narrowing of the calibration method, not a full reproduction of the rebuttal paper's 13-metric framework. Geneformer (zero-shot) outputs a per-cell cosine-shift ranking, not a predicted expression profile, so it is excluded from the table above and reported in its own ranking table in the full write-up.